# Dikaios Amanah

A Secure, Transparent and Ethical SME Investment Platform Based on Musharakah Principles




## Project Proposal

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# 1. Executive Summary

Dikaios Amanah is a Software Engineering capstone project that aims to bridge the financing gap faced by Small and Medium Enterprises (SMEs) in Bangladesh through a secure, transparent, and ethical digital investment platform. The proposed system is inspired by the Islamic finance principle of Musharakah, where investors and business owners share profits and losses instead of relying on interest-based lending.

Many SMEs struggle to obtain financing due to collateral requirements, limited access to venture capital, and a lack of trustworthy investment mechanisms. At the same time, many individual investors seek ethical investment opportunities that align with Islamic financial principles while offering greater transparency and accountability. Existing financing solutions rarely address the needs of both groups simultaneously.

Dikaios Amanah provides a digital platform that connects verified SMEs with verified retail investors through a structured investment process. Businesses submit funding requests, supporting documents, and business information, while investors browse verified investment opportunities and participate according to their financial capacity. The platform focuses on transparency, risk awareness, and secure information management rather than acting as a financial institution.

To enhance trust, the system incorporates a Permissioned Trust and Audit Layer that records critical events such as business verification, contract approvals, investment agreements, and profit declarations in a tamper-evident manner. Operational data is managed using a relational database, while cryptographic hashing and immutable audit records help maintain data integrity without replacing conventional database systems.

The proposed system will be implemented as a Minimum Viable Product (MVP) using modern web technologies including React, Node.js, Express.js, PostgreSQL, Firebase Storage, JWT authentication, bcrypt password hashing, and SHA-256 document hashing. Advanced features such as live payment gateway integration, regulatory compliance, and production-scale blockchain deployment are intentionally excluded from the MVP and considered future enhancements. This project demonstrates the practical application of software engineering principles, secure system design, database management, authentication mechanisms, and ethical financial technology while addressing a real-world problem with a technically feasible solution.

# 2. Project Background

The Small and Medium Enterprise (SME) sector is widely recognized as one of the primary drivers of economic growth, employment generation, and innovation in developing countries. In Bangladesh, SMEs contribute significantly to industrial production, regional economic development, and job creation. Despite their importance, many promising businesses struggle to obtain sufficient financial support during their early and growth stages.

Traditional financing methods primarily rely on collateral-based lending, where businesses must provide valuable assets before obtaining loans. Many small business owners, particularly new entrepreneurs, are unable to meet these collateral requirements despite having viable business ideas and sustainable business models. As a result, a large number of SMEs remain financially underserved.

At the same time, many individual investors seek opportunities to invest their savings in productive businesses. However, existing investment channels often require substantial capital, lack transparency, or operate under financial models that may not align with the ethical or religious preferences of many investors in Bangladesh. This creates a disconnect between businesses requiring capital and individuals willing to invest.

Islamic finance offers several partnership-based investment models that emphasize fairness, transparency, and shared responsibility. Among these models, Musharakah represents a partnership where all participating parties contribute capital and share both profits and losses according to mutually agreed terms. Unlike interest-based lending, Musharakah encourages risk sharing, cooperation, and long-term business growth.

With the rapid adoption of digital financial technologies, secure online platforms have become increasingly capable of connecting multiple stakeholders while maintaining transparency and accountability. Modern Software Engineering practices, cloud computing, secure authentication, cryptographic techniques, and digital document management make it possible to build trustworthy investment platforms capable of supporting complex business interactions.

Recent advancements in web technologies have significantly improved the feasibility of developing scalable and secure financial platforms. Technologies such as React, Node.js, PostgreSQL, Firebase Storage, and modern authentication mechanisms enable developers to build responsive and reliable applications while maintaining strong security standards.

Despite these technological advancements, there remains a shortage of platforms specifically designed for ethical partnership-based investments targeting verified local SMEs. Most existing crowdfunding platforms primarily support donations, reward-based campaigns, or startup fundraising, while traditional banking systems continue to focus on debt financing rather than partnership models.

Dikaios Amanah aims to address this gap by providing a secure digital ecosystem that connects verified SMEs with verified investors through a transparent Musharakah-based investment process. Rather than replacing existing financial institutions, the proposed platform serves as a technology-enabled intermediary that improves trust, documentation, transparency, and accountability throughout the investment lifecycle.

The project also demonstrates the practical application of Software Engineering principles in solving a real-world financial problem. It integrates modern web development, secure authentication, database management, document verification, cryptographic hashing, and a Permissioned Trust and Audit Layer to create a technically feasible Minimum Viable Product (MVP) suitable for academic development while remaining scalable for future enhancements.

# 3. Problem Statement

## 3.1 Overview

Although Bangladesh has experienced significant economic growth over the past decade, access to business financing remains a major challenge for many Small and Medium Enterprises (SMEs). Entrepreneurs frequently struggle to secure capital for business expansion, equipment purchases, inventory management, and operational growth. While financing opportunities exist through commercial banks and financial institutions, many SMEs remain excluded due to strict eligibility requirements and limited access to ethical investment alternatives.

At the same time, many individuals possess savings that could be invested in productive businesses. However, they often lack access to transparent, trustworthy, and ethically governed investment platforms. This creates a disconnect between businesses seeking capital and investors seeking responsible investment opportunities.

The following subsections describe the primary problems addressed by the proposed

system.

## 3.2 Challenges Faced by SMEs

Small businesses encounter numerous barriers when attempting to obtain external financing. Traditional financial institutions generally require collateral, lengthy approval procedures, financial history, and extensive documentation before approving loans. New entrepreneurs and micro-businesses often fail to satisfy these requirements despite having commercially viable business ideas.

Furthermore, financing through conventional lending introduces fixed repayment obligations regardless of business performance. This increases financial pressure during periods of low revenue and may discourage entrepreneurs from pursuing business expansion.

Major SME challenges include:

- Difficulty obtaining financing without collateral.
- Limited visibility to potential investors.
- Lengthy loan approval procedures.
- Dependence on interest-based financing.
- Limited access to ethical partnership-based investment models.

## 3.3 Challenges Faced by Investors

Retail investors also encounter significant difficulties when attempting to invest directly in legitimate businesses. Existing investment opportunities frequently require substantial capital, provide limited transparency, or offer insufficient information regarding business performance and associated risks.

Many investors are interested in ethical investment opportunities that comply with Islamic financial principles. However, reliable digital platforms supporting partnership-based investment remain extremely limited.

Major investor challenges include:

- Lack of verified investment opportunities.

- Limited transparency regarding business information.
- Difficulty evaluating investment risk.
- Fear of fraud and document manipulation.
- Limited access to Shariah-compliant investment options.

## 3.4 Limitations of Existing Solutions

Current financing methods only partially address the needs of SMEs and investors.

Commercial banks primarily focus on collateral-based lending rather than partnership investment. Venture capital firms generally invest only in high-growth startups capable of generating significant returns, leaving many small businesses underserved. Conventional crowdfunding platforms often emphasize donations or reward-based campaigns instead of structured investment partnerships.

Moreover, most existing digital platforms provide limited mechanisms for document verification, auditability, and long-term transparency throughout the investment lifecycle.

## 3.5 Trust and Transparency Issues

Trust represents one of the most critical barriers preventing direct investment between businesses and individuals.

Investors require confidence that businesses provide authentic documentation, accurate financial information, and transparent reporting. Likewise, businesses require assurance that investors are legitimate participants capable of fulfilling agreed financial commitments.

Without reliable verification mechanisms, secure authentication, immutable audit records, and transparent documentation, both parties remain exposed to fraud, misinformation, and contractual disputes.

The absence of an independent audit mechanism further increases concerns regarding document tampering, unauthorized record modification, and accountability.

### 3.6 Need for a Software-Based Solution

The problems identified above demonstrate the need for a secure digital platform capable of connecting verified SMEs and verified investors within a transparent and ethically governed ecosystem.

Rather than replacing financial institutions, the proposed platform focuses on improving trust, transparency, documentation, verification, and investment management through modern Software Engineering practices.

The proposed system incorporates secure authentication, role-based access control, document verification, encrypted storage, cryptographic hashing, and a Permissioned Trust and Audit Layer to strengthen confidence among all stakeholders while maintaining a technically feasible architecture suitable for a university-level Minimum Viable Product (MVP).

## 4. Proposed Solution

### 4.1 Overview

Dikaios Amanah is proposed as a secure web-based investment platform designed to connect verified Small and Medium Enterprises (SMEs) with verified retail investors through a transparent and ethically governed investment ecosystem. The platform follows the Islamic finance principle of Musharakah, where participating parties share both profits and losses based on mutually agreed terms rather than fixed interest payments.

Unlike traditional crowdfunding systems, Dikaios Amanah focuses on building trust through structured verification, secure document management, transparent reporting, and a Permissioned Trust and Audit Layer. The proposed system acts as a technology-enabled intermediary that facilitates responsible investment while improving accountability throughout the investment lifecycle.

### 4.2 System Overview

The platform consists of three primary stakeholders:

- Business Owners

- Investors
- System Administrators

Each stakeholder interacts with the system according to predefined roles and permissions enforced through secure authentication and authorization mechanisms.

The proposed system provides separate dashboards tailored to each user category, ensuring that users only access functions relevant to their responsibilities.

## 4.3 Business Owner Module

Business owners use the platform to seek investment for business expansion.

The module provides the following capabilities:

- Business registration
- Identity verification
- Business profile management
- Secure document upload
- Investment campaign creation
- Progress reporting
- Profit declaration
- Investor communication

Before becoming visible to investors, every business must successfully complete verification and administrative review.

## 4.4 Investor Module

The investor module enables individuals to discover verified investment opportunities while maintaining transparency throughout the investment process.

Major features include:

- Investor registration

- Know Your Customer (KYC) verification
- Business discovery
- Risk score review
- Investment management
- Portfolio monitoring
- Profit tracking
- Investment history

Investors can evaluate businesses using verified information before making investment decisions.

## 4.5 Administrator Module

The administrator oversees platform integrity and operational management.

Responsibilities include:

- User verification
- Business approval
- Document validation
- Risk assessment management
- Campaign moderation
- Audit monitoring
- User management
- Platform reporting

Administrative actions are recorded to improve accountability and transparency.

## 4.6 System Workflow

The proposed investment workflow consists of the following stages:

1. Business registration.
2. Identity and document verification.
3. Risk assessment.
4. Administrative approval.
5. Campaign publication.
6. Investor review and funding.
7. Musharakah agreement confirmation.
8. Business operation and progress reporting.
9. Profit declaration.
10. Profit distribution.
11. Campaign completion.

This structured workflow ensures that all investment activities are documented and performed in a transparent manner.

## 4.7 Security Mechanisms

Security forms one of the fundamental design principles of Dikaios Amanah.

The proposed platform incorporates multiple layers of protection, including:

- JWT-based authentication
- Role-Based Access Control (RBAC)
- Secure password hashing using bcrypt
- SHA-256 document hashing

- Secure cloud document storage
- Input validation
- Activity logging
- Session management

Rather than relying on a single security mechanism, the system adopts a layered security approach to protect user accounts, business information, and investment records.

## 4.8 Permissioned Trust and Audit Layer

To strengthen trust between stakeholders, the proposed platform incorporates a Permissioned Trust and Audit Layer.

Instead of storing operational business data on a blockchain, the system records cryptographic hashes and timestamps of critical events, including:

- Business verification
- Contract approval
- Investment confirmation
- Profit declaration
- Audit records
- Document integrity verification

The blockchain functions as an immutable audit mechanism, while PostgreSQL remains responsible for day-to-day application data. This separation provides transparency and tamper-evident verification without compromising system performance.

## 4.9 Expected Outcome

The proposed platform establishes a secure and transparent digital ecosystem that enables SMEs to access ethical investment while allowing investors to participate confidently through verified business information and structured partnership agreements.

By combining modern web technologies with secure software engineering practices, Dikaios Amanah aims to demonstrate a practical and scalable solution for ethical SME investment that is achievable as a university-level Minimum Viable Product while providing a strong foundation for future commercial deployment.

# 5. System Architecture

## 5.1 Overview

The proposed system follows a modern three-tier web application architecture consisting of a presentation layer, an application layer, and a data layer. This architecture promotes modularity, maintainability, scalability, and security by separating user interaction, business logic, and persistent data storage.

The architecture has been designed to support future expansion while remaining technically feasible for implementation as a university-level Minimum Viable Product (MVP).

## 5.2 Presentation Layer

The Presentation Layer provides the graphical user interface through which users interact with the platform.

The frontend will be developed using React.js and consists of three dedicated user portals:

- Business Owner Portal
- Investor Portal
- Administrator Dashboard

The interface communicates with backend services using RESTful APIs over HTTPS.

## 5.3 Application Layer

The Application Layer contains the business logic responsible for processing user requests and enforcing business rules.

This layer will be implemented using Node.js and Express.js.

Major responsibilities include:

- User Authentication
- Authorization
- Business Verification
- Investment Management
- Risk Assessment
- Profit Distribution Logic
- Document Management
- Audit Management

## 5.4 Data Layer

The Data Layer stores all operational data required by the platform.

PostgreSQL serves as the primary relational database for users, businesses, investments, contracts, and transaction records.

Firebase Storage stores uploaded business documents and supporting files.

The Permissioned Trust and Audit Layer stores cryptographic hashes and timestamps of critical platform events to provide tamper-evident audit records.

## 5.5 Security Architecture

Security is implemented through multiple independent layers.

- HTTPS Communication
- JWT Authentication
- Role-Based Access Control
- bcrypt Password Hashing
- SHA-256 Document Hashing
- Secure File Storage

- Audit Logging
- Permissioned Trust and Audit Layer

This layered security model minimizes the impact of individual security failures while improving overall system resilience.

### 5.6 Architectural Benefits

The proposed architecture provides several advantages.

- Separation of Concerns
- Modular Development
- Scalability
- Improved Security
- Easier Maintenance
- Future Cloud Deployment
- Support for Additional Services

## 6. Project Objectives

### 6.1 General Objective

The primary objective of Dikaios Amanah is to design and develop a secure, transparent, and ethically governed web-based investment platform that enables verified Small and Medium Enterprises (SMEs) in Bangladesh to connect with verified retail investors through Musharakah-based partnership financing. The project aims to demonstrate the practical application of modern Software Engineering principles while addressing real-world challenges related to transparency, trust, security, and ethical investment.

## 6.2 Specific Objectives

The project seeks to achieve the following objectives:

1. Develop a responsive web application that supports separate interfaces for business owners, investors, and administrators.
2. Implement secure user authentication and role-based authorization to protect system resources and user accounts.
3. Design a structured business verification process that improves investor confidence before investment decisions are made.
4. Provide a secure document management system for storing and managing business verification documents.
5. Enable SMEs to create, manage, and monitor Musharakah-based investment campaigns.
6. Allow investors to discover verified businesses, review investment opportunities, and monitor their investment portfolio.
7. Record critical platform events using a Permissioned Trust and Audit Layer to improve transparency and provide tamper-evident audit records.
8. Apply secure software engineering practices, including password hashing, document integrity verification, secure session management, and activity logging.
9. Develop a technically feasible Minimum Viable Product (MVP) suitable for demonstration as a final-year Software Engineering capstone project.

## 6.3 Expected Deliverables

Upon successful completion, the project is expected to deliver:

- A fully functional web-based MVP.
- Separate dashboards for Business Owners, Investors, and Administrators.
- Business registration and verification module.

- Investor registration and KYC module.
- Investment campaign management system.
- Risk assessment and approval workflow.
- Secure authentication and authorization system.
- Permissioned Trust and Audit Layer prototype.
- Complete Software Requirements Specification (SRS).
- UML diagrams and software design documentation.
- User documentation and project report.

### 6.4 Success Criteria

The proposed project will be considered successful if it is able to:

- Successfully authenticate and authorize different categories of users.
- Enable verified businesses to publish investment campaigns.
- Allow investors to browse and participate in verified investment opportunities.
- Maintain secure storage and verification of business documents.
- Demonstrate transparent recording of critical events through the Permissioned Trust and Audit Layer.
- Provide an intuitive and responsive user interface.
- Demonstrate compliance with the defined functional and non-functional requirements of the proposed system.

## 7. Project Scope

### 7.1 Overview

The scope of Dikaios Amanah defines the boundaries of the proposed system and identifies the features that will be included in the Minimum Viable Product (MVP). The project

focuses on designing and developing a secure, web-based platform that enables transparent and ethical investment between verified Small and Medium Enterprises (SMEs) and verified retail investors using the Musharakah partnership model.

The project emphasizes Software Engineering principles, secure system design, role-based access control, document verification, and transparent audit mechanisms while remaining technically feasible for implementation as a final-year capstone project.

## 7.2 Functional Scope

The proposed system will include the following functional modules:

- User Registration and Authentication
- Business Owner Dashboard
- Investor Dashboard
- Administrator Dashboard
- Business Profile Management
- Investor KYC Verification
- Business Verification
- Secure Document Upload
- Investment Campaign Creation
- Campaign Browsing
- Investment Tracking
- Profit Declaration
- Portfolio Management
- Notification System
- Activity Logging
- Permissioned Trust and Audit Layer

## 7.3 Technical Scope

The project will be developed using modern web technologies and secure software engineering practices.

The planned technology stack includes:

- React.js for the frontend.
- Node.js and Express.js for backend services.
- PostgreSQL for relational database management.
- Firebase Storage for secure document storage.
- JSON Web Token (JWT) for authentication.
- bcrypt for password hashing.
- SHA-256 for document integrity verification.
- Git and GitHub for version control.
- A Permissioned Trust and Audit Layer for recording critical system events.

The proposed architecture follows a layered design that separates presentation, business logic, security, and data management to improve maintainability and scalability.

## 7.4 Project Stakeholders

The proposed system involves three primary stakeholder groups.

1. Business Owners

   Businesses seeking ethical investment opportunities for expansion and operational growth.

2. Retail Investors

   Individuals interested in participating in verified Musharakah-based investment opportunities.

3. System Administrators

   Authorized personnel responsible for user verification, business approval, system monitoring, and platform administration.

## 7.5 Project Limitations

To maintain feasibility within the academic timeframe, several features are intentionally excluded from the MVP.

The proposed system does not include:

- Live banking integration.
- Real payment gateway integration.
- BSEC regulatory compliance implementation.
- AI-based fraud detection.
- AI-powered risk prediction.
- Production Hyperledger Fabric deployment.
- Native Android or iOS mobile applications.
- Automated smart contract execution.

These features are identified as future enhancements rather than core project requirements.

## 7.6 Project Assumptions

The development of the proposed system is based on the following assumptions.

- Users possess basic digital literacy.
- Internet connectivity is available.
- Businesses provide authentic documentation.
- Administrators perform verification responsibly.

- The project is developed for academic demonstration purposes.
- Simulated investment transactions are used during system testing.

### 7.7 Future Scope

Although the MVP focuses on demonstrating the complete investment workflow, the proposed architecture has been designed to support future expansion.

Potential future enhancements include:

- Mobile application development.
- Integration with digital payment gateways.
- Hyperledger Fabric deployment.
- AI-assisted business risk assessment.
- Automated regulatory reporting.
- Smart contract automation.
- Investor recommendation system.
- Multi-language support.
- Real-time analytics dashboard.
- Integration with government business verification services.

These enhancements would enable Dikaios Amanah to evolve into a production-ready ethical investment platform while maintaining compatibility with the proposed system architecture.

## 8. Feasibility Study

### 8.1 Overview

A feasibility study evaluates whether the proposed system can be successfully developed within the available resources, time, technical knowledge, and budget. The objective is to

determine whether Dikaios Amanah is practical as a university-level Software Engineering capstone project while also demonstrating the potential for future real-world deployment.

The proposed system has been designed as a Minimum Viable Product (MVP), ensuring that essential features can be implemented within the academic timeline while allowing future expansion beyond the scope of this project.

## 8.2 Technical Feasibility

The proposed system utilizes mature and widely adopted web development technologies that are well supported by extensive documentation and open-source communities.

The planned technology stack consists of:

| **Technology** | **Purpose** |
|---|---|
| React.js | Frontend development |
| Node.js | Backend runtime environment |
| Express.js | REST API development |
| PostgreSQL | Relational database management |
| Firebase Storage | Secure document storage |
| JWT | User authentication |
| bcrypt | Password hashing |
| SHA-256 | Document integrity verification |
| Git & GitHub | Version control |
| Permissioned Trust and Audit Layer | Tamper-evident audit records |

Table 1: Proposed Technology Stack

These technologies are suitable for building secure, scalable, and maintainable web applications. Their widespread adoption also ensures long-term maintainability and extensive learning resources.

The proposed layered architecture separates presentation, business logic, security, and data management, reducing development complexity and improving software maintainability.

Therefore, the project is considered technically feasible.

## 8.3 Economic Feasibility

The project is intended for academic demonstration and therefore minimizes operational costs by utilizing free and open-source software wherever possible.

The estimated development cost is limited primarily to internet access, optional cloud hosting, domain registration, and testing resources.

Open-source technologies such as React, Node.js, PostgreSQL, Express.js, and GitHub significantly reduce software licensing costs.

Cloud services such as Firebase Storage also provide generous free usage tiers suitable for MVP development.

Consequently, the project can be completed with minimal financial investment.

## 8.4 Operational Feasibility

The proposed platform has been designed with usability and accessibility in mind.

Business owners will be able to submit investment campaigns through an intuitive dashboard, while investors will have access to verified business information before making investment decisions.

System administrators will manage verification, approvals, and platform monitoring through a dedicated administrative interface.

Role-Based Access Control (RBAC), secure authentication, activity logging, and document verification improve operational security while simplifying day-to-day platform management.

The system architecture supports future scalability without requiring major architectural changes.

Therefore, the project is considered operationally feasible.

## 8.5 Schedule Feasibility

The proposed project has been divided into manageable development phases following an iterative Software Development Life Cycle (SDLC).

Major development phases include:

- Requirements Analysis
- System Design
- Database Design
- UI/UX Design

- Backend Development
- Frontend Development
- Security Implementation
- Trust and Audit Layer Development
- Testing and Debugging
- Documentation

The MVP intentionally excludes complex production-level features such as live banking integration, regulatory compliance, and smart contract automation to ensure successful completion within the academic schedule.

Based on the planned scope and available development timeline, the project is considered schedule feasible.

## 8.6 Risk Assessment

Although the project is considered feasible, several risks have been identified.

| **Risk** | **Impact** | **Mitigation Strategy** |
|---|---|---|
| Development Time Constraints | Medium | Prioritize MVP features. |
| Learning New Technologies | Medium | Incremental learning and prototyping. |
| Authentication Security Issues | High | Follow secure coding practices and testing. |
| Database Design Errors | Medium | Early ERD design and normalization. |
| Document Verification Challenges | Medium | Administrative review process. |
| Blockchain Integration Complexity | Medium | Implement a lightweight permissioned audit layer for the MVP. |

Table 2: Project Risk Assessment

Proper planning, iterative development, version control, and continuous testing will significantly reduce these risks throughout the software development lifecycle.

### 8.7 Feasibility Conclusion

Based on the technical, economic, operational, and schedule analyses, Dikaios Amanah is considered a feasible Software Engineering capstone project. The selected technology stack, clearly defined MVP scope, and phased development strategy provide a realistic pathway for successful implementation within the available academic timeframe while allowing future enhancements for production deployment.

## 9. Work Breakdown Structure (WBS)

### 9.1 Overview

The Work Breakdown Structure (WBS) is a hierarchical decomposition of the project into manageable tasks and deliverables. It provides a structured roadmap for planning, monitoring, and controlling the development process. Each major phase is divided into smaller work packages that can be scheduled, implemented, and evaluated independently.

The WBS for Dikaios Amanah follows the Software Development Life Cycle (SDLC) and ensures that every aspect of the project is systematically planned.

### 9.2 Level 1 Work Breakdown Structure

| WBS ID | Major Phase |
|---|---|
| 1.0 | Project Initiation |
| 2.0 | Requirements Analysis |
| 3.0 | System Design |
| 4.0 | Database Design |
| 5.0 | Frontend Development |
| 6.0 | Backend Development |
| 7.0 | Security Implementation |
| 8.0 | Trust and Audit Layer Development |
| 9.0 | Testing and Quality Assurance |
| 10.0 | Documentation |
| 11.0 | Final Deployment and Demonstration |

Table 3: Level 1 Work Breakdown Structure

### 9.3 Detailed Work Breakdown Structure

| **WBS ID** | **Task Description** |
|---|---|
| 1.1 | Literature Review |
| 1.2 | Problem Identification |
| 1.3 | Project Proposal Preparation |
| 1.4 | Requirement Gathering |
| 2.1 | Functional Requirement Analysis |
| 2.2 | Non-functional Requirement Analysis |
| 2.3 | Stakeholder Identification |
| 2.4 | Use Case Analysis |
| 3.1 | System Architecture Design |
| 3.2 | UI/UX Wireframe Design |
| 3.3 | UML Diagram Preparation |
| 3.4 | API Design |
| 4.1 | ER Diagram Design |
| 4.2 | Database Schema Design |
| 4.3 | Database Normalization |
| 5.1 | Authentication Pages |
| 5.2 | Business Owner Dashboard |
| 5.3 | Investor Dashboard |
| 5.4 | Administrator Dashboard |
| 5.5 | Investment Campaign Interface |
| 6.1 | Authentication API |
| 6.2 | Business Management API |
| 6.3 | Investment Management API |
| 6.4 | Document Management API |
| 6.5 | Reporting API |
| 7.1 | JWT Authentication |
| 7.2 | Password Hashing (bcrypt) |
| 7.3 | Role-Based Access Control |
| 7.4 | SHA-256 Document Hashing |
| 7.5 | Activity Logging |

| | |
|---|---|
| 8.1 | Audit Layer Design |
| 8.2 | Hash Generation Module |
| 8.3 | Immutable Audit Record Storage |
| 9.1 | Unit Testing |
| 9.2 | Integration Testing |
| 9.3 | System Testing |
| 9.4 | Bug Fixing |
| 10.1 | Technical Documentation |
| 10.2 | User Manual |
| 10.3 | Final Report Writing |
| 11.1 | Final Demonstration |
| 11.2 | Project Submission |

### 9.4 WBS Summary

The proposed Work Breakdown Structure divides the project into manageable development phases that align with the Software Development Life Cycle. Each work package represents a clearly defined deliverable, allowing systematic project management, progress monitoring, and resource allocation throughout the development process. The structured approach also minimizes project risks by enabling incremental development and continuous testing.

## 10. Project Timeline

### 10.1 Overview

A project timeline provides a structured schedule for completing all planned activities within the academic semester. The timeline follows an iterative Software Development Life Cycle (SDLC), allowing each phase to build upon the previous one while providing sufficient time for testing, documentation, and revisions.

Since Dikaios Amanah is being developed as a solo capstone project, the development schedule prioritizes the implementation of core MVP features before optional en-

hancements. This approach reduces project risk and ensures that the essential system components are completed within the available timeframe.

## 10.2 Project Timeline

| **Activity** | **Duration** |
|---|---|
| Literature Review | Week 1 |
| Requirement Analysis | Week 2 |
| Project Proposal Preparation | Week 2 |
| System Design | Week 3 |
| Database Design | Week 3 |
| UI/UX Design | Week 4 |
| Frontend Development | Weeks 5–7 |
| Backend Development | Weeks 6–9 |
| Authentication and Security Implementation | Week 8 |
| Trust and Audit Layer Development | Week 9 |
| System Integration | Week 10 |
| Testing and Debugging | Weeks 11–12 |
| Documentation | Weeks 13–14 |
| Final Improvements | Week 15 |
| Final Presentation and Submission | Week 16 |

Table 5: Proposed Project Timeline

## 10.3 Milestones

The project is expected to achieve the following milestones during development.

- Approval of the project proposal.
- Completion of the Software Requirements Specification (SRS).
- Completion of system architecture and database design.
- Development of frontend and backend modules.
- Integration of security mechanisms.
- Implementation of the Permissioned Trust and Audit Layer.
- Completion of testing and debugging.
- Final project demonstration.
- Submission of project documentation.

### 10.4 Timeline Summary

The proposed timeline allocates sufficient time for planning, development, testing, documentation, and presentation while maintaining a realistic workload for a solo developer. The phased schedule supports incremental development and allows continuous evaluation of project progress throughout the Software Development Life Cycle.

## 11. Budget Estimation

### 11.1 Overview

The budget estimation represents the approximate cost required to establish and operate Dikaios Amanah as a functional digital investment platform beyond the prototype stage. Existing development hardware and open-source software are assumed to be available and are therefore excluded from the estimation. The budget focuses on infrastructure, deployment, legal compliance, business registration, security, and operational expenses. Market prices are based on publicly available information from relevant vendors and service providers.

## 11.2 Estimated Project Budget

| **Item** | **Quantity** | **Estimated Cost (BDT)** | **Reference Source** |
|---|---|---|---|
| Business Registration | 1 | 10,000 | RJSC Bangladesh |
| Trade License | 1 | 5,000 | Local City Corporation |
| Professional Legal Consultation | 1 | 20,000 | Local Legal Firms |
| Domain Name (.com, 1 Year) | 1 | 1,500 | Namecheap |
| Cloud VPS (Basic VPS - 12 Months) | 1 | 7,900 | Hostinger VPS |
| SSL Certificate (if required) | 1 | 5,000 | DigiCert / Sectigo |
| Email Hosting (Business Email) | 1 Year | 6,000 | Google Workspace |
| Identity Verification (KYC Integration)* | Initial Setup | 25,000 | Estimated API Integration Cost |
| Security Audit and Penetration Testing | 1 | 30,000 | Local Cybersecurity Service Providers |
| Contingency (5%) | – | 5,525 | Estimated |
| **Total Estimated Cost** | | **115,925 BDT** | |

Table 6: Estimated Budget for Initial Platform Deployment

*Estimated integration cost for connecting to a third-party identity verification or KYC service. The MVP will simulate this functionality and will not integrate with live providers.

## 11.3 Budget Justification

The proposed budget reflects the estimated expenditure required to transition the platform from an academic prototype to an operational software service. The largest expenses relate to legal compliance, business registration, cybersecurity, identity verification, and cloud infrastructure, which are essential for operating an online investment platform.

The MVP developed as part of this capstone project will not incur all of these expenses because it uses open-source technologies and simulated services. However, these estimated costs demonstrate the financial considerations involved in commercializing the proposed platform and are supported by publicly available pricing from relevant vendors and service providers.

The use of open-source technologies including React.js, Node.js, Express.js, PostgreSQL, Git, and Visual Studio Code eliminates software licensing fees while maintaining a modern and scalable technology stack.

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